

Project Performance Testing

Functional & Performance Testing

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Testing:

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass / Fail
FT-01	Valid input prediction	Enter N=90, P=42, K=43, temp=21, humidity=82, pH=6.5, rainfall=202 and click Recommend Crop	System displays 'Rice' with confidence score	Correctly displayed 'Rice' with 97% confidence	Pass
FT-02	Low rainfall input prediction	Enter N=20, P=68, K=20, temp=29, humidity=24, pH=6.0, rainfall=65 and click Recommend Crop	System recommends a drought-tolerant crop	Correctly recommended 'Kidneybeans'	Pass
FT-03	Top-3 alternate suggestions	After any valid prediction, check the alternate suggestions section	3 alternate crops shown with confidence percentages	Top-3 crops displayed correctly below main recommendation	Pass
FT-04	Empty/blank input	Leave a field empty and click Recommend Crop	Clear error message shown, app does not crash	Error message displayed, no crash occurred	Pass

FT-05	Model load check	Verify app loads correctly with model.pkl present	App loads without errors, form is visible	App loaded correctly and form displayed	Pass
FT-06	Error handling for wrong values	Enter a text character in a numeric field	Clear error message shown to user	Handled gracefully with readable error message	Pass

Performance Testing:

Test Case ID	Performance Metric	Target	Actual	Pass / Fail
PT-01	Model Accuracy on Test Data (20% split)	≥ 90%	99.32%	Pass
PT-02	Prediction Response Time	< 3 seconds	~1 second	Pass
PT-03	App Load Time (deployed link)	< 5 seconds	~3 seconds	Pass
PT-04	App accessibility on mobile browser	Should work on mobile	Tested on phone browser - fully functional	Pass